

Tejas Patil

Davis, CA

tpatil@ucdavis.edu | +1 (530) 979-6396 | tejassvpatil.github.io | [LinkedIn](#) | [GitHub](#) | [Google Scholar](#)

Research Profile

PhD student in Computer Science at UC Davis working at the intersection of artificial intelligence, healthcare, and human-centered sensing. Research focuses on multimodal machine learning and digital-health systems for cognitive health, remote care, caregiver support, and real-world clinical and life-science applications.

Education

University of California, Davis

PhD, Computer Science; GPA: 4.0

Sep 2023–present

Davis, CA

Advisors: Alyssa Weakley and Chen-Nee Chuah.

Relevant coursework: AI for Health; Vision-Language Modeling; Advanced Deep Learning; Machine Learning and Discovery; Algorithm Design and Analysis.

University of Pune

BE, Computer Engineering; GPA: 3.74

Jul 2017–Jul 2021

Pune, India

Relevant coursework: Machine Learning; Discrete Mathematics; Artificial Intelligence and Robotics; Data Structures and Algorithms; Theory of Computation; Computer Networks.

Current Position

UC Davis Health — Graduate Student Researcher

Department of Neurology, Everyday Cognition and Resilient Aging Lab

Mar 2025–present

Davis, CA

Research areas: AI for healthcare, human-centered systems, multimodal sensing, remote caregiving, health technology, and software systems for clinical research.

Research Experience

UC Davis Health — Graduate Student Researcher

Department of Neurology

Mar 2025–present

Davis, CA

- Integrated in-home sensing streams into the I-Care caregiver platform by defining sampling strategies, data contracts, sensor-derived activity summaries, privacy-aware visualizations, and access-control boundaries for sensitive health-adjacent data.
- Built real-time caregiver-patient workflows including one-to-one messaging, media sharing, and calendar synchronization, iterating on usability for cognitively accessible interaction.
- Optimized Firestore data models and access patterns across events and permissions, reducing redundant reads and writes and improving end-to-end responsiveness by approximately 30% through schema refactors, query consolidation, indexing, and tighter authorization boundaries.
- Developed backend services and APIs using Firebase, FastAPI, Python, JavaScript, React, and Vue to support remote caregiving, patient-caregiver communication, and clinical research workflows.

University of Pune — Undergraduate Researcher

Computer Engineering

Dec 2019–Jun 2021

Pune, India

- Designed and evaluated a Zero-Knowledge Proof-based authentication system using Secure Remote Password and AES to reduce exposure of plaintext credentials and strengthen resistance to replay, man-in-the-middle, sniffing, and offline guessing attacks.
- Performed survey and bibliometric research on Zero-Knowledge Proofs for authentication, analyzing Scopus-indexed literature from 2010–2021, publication trends, geographic contributions, keywords, citation patterns, and applications including IoT and wireless body-area networks.
- Co-authored three papers: an IEEE ASIANCON conference paper, a Library Philosophy and Practice bibliometric article, and a survey of IoT security methodologies.

Professional Experience

ABN AMRO Bank N.V. — Software Engineer / Associate Consultant
Capgemini

Aug 2021–Aug 2023
India / Remote

- Designed and implemented Java backend services and REST APIs for commercial-banking workflows, scalable customer operations, and digital onboarding systems.
- Built and optimized API-driven customer workflows, improving key conversion metrics by 60% and reducing Customer Effort Score by 40%.
- Delivered features across the Agile software-development lifecycle; maintained Azure CI/CD pipelines, wrote unit and integration tests, debugged production issues using logs and metrics, and collaborated on requirements and technical documentation.

Capgemini — Software Engineer Intern
Engineering Internship

Feb 2021–Apr 2021
India

- Developed a sports-equipment e-commerce prototype using Spring Boot and Angular, including backend services and REST APIs for product and order workflows.
- Designed and managed Oracle 11g relational-database operations for efficient data access and transaction integrity.

Peer-Reviewed Publications

1. P. Koppolu, A. Gohil, **T. Patil**, P. Yadav, M. Cohn, C.-N. Chuah, S. Pan, S. Tomaszewski Farias, and A. Weakley. “Streamlining Remote Care: Computer Access Solutions for Older Adults with Cognitive Impairment.” *AHFE International Conference*, 2026.
2. A. Gohil, D. Bissig, H. Wang, P. Koppolu, **T. Patil**, P. Yadav, S. Pan, M. Cohn, C.-N. Chuah, S. Tomaszewski Farias, and A. Weakley. “Embedding Self-Administered Dementia Monitoring into a Remote Caregiving Platform: A Pilot Usability and Data-Sharing Preferences Study.” *AHFE International Conference*, 2026.
3. **T. Patil**, P. Koppolu, A. Gohil, P. Yadav, C.-N. Chuah, S. Tomaszewski Farias, and A. Weakley. “Evaluating Integrated Multimedia Communication Designed for Cognitive Accessibility within a Remote Caregiving Platform.” *AHFE International Conference*, 2026.
4. P. Yadav, H. Agarwal, P. Koppolu, A. Gohil, **T. Patil**, C.-N. Chuah, S. Pan, S. Aswar, S. Pimento, M. Cohn, and A. Weakley. “Bounded Delegation of a Large Language Model in a Conversational Agent for Dementia Support.” *IEEE AI x Medical, Health, and Care (AIxMHC)*, 2026.
5. D. Y. Lee, **T. Patil**, S. Pan, and A. Weakley. “Only Speaking FACTS: Foundation Models for Activity-Centric Timeline Description.” *HumanSys*, 2026.
6. D. Y. Lee, A. Weakley, **T. Patil**, H. Wei, D. Cardona, and S. Pan. “AlzGood: Validating Remote Care Systems in the Wild via an Interactive Deployment Framework.” *IEEE Connected Health: Applications, Systems and Engineering Technologies (CHASE)*, 2026.
7. P. Yadav, M. Cohn, P. Koppolu, H. Agarwal, A. Gohil, **T. Patil**, S. Pimento, and A. Weakley. “AI-Care: A Conversational Agentic System for Task Coordination in Alzheimer’s Disease Care.” *IEEE Connected Health: Applications, Systems and Engineering Technologies (CHASE)*, 2026.
8. A. Weakley, S. Pimento, A. Gohil, P. Hegde, A. Kaur, P. Koppolu, H. Agarwal, P. Yadav, **T. Patil**, A. Weakley, and S. Tomaszewski Farias. “[Promoting Autonomy in Older Adults with Cognitive Impairment: Co-Designing an Interactive Calendar for Memory Support](#).” *Human Factors in Design, Engineering, and Computing*, AHFE Open Access, vol. 199, 2025.
9. A. Pathak, **T. Patil**, S. Pawar, P. Raut, and S. Khairnar. “[Secure Authentication Using Zero Knowledge Proof](#).” *IEEE ASIANCON*, 2021.
10. A. Pathak, **T. Patil**, S. Pawar, P. Raut, S. Khairnar, and S. Gite. “[Bibliometric Survey on Zero-Knowledge Proof for Authentication](#).” *Library Philosophy and Practice*, 2021.
11. A. Pathak, **T. Patil**, S. Pawar, P. Raut, and S. Khairnar. “[A Survey on Methodologies for Intensifying the Security in IoT Environment](#).” *Journal of Critical Reviews*, 7(19), 1135–1147, 2020.

Invited Symposium Presentation

S. Pimento, H. Agarwal, A. Gohil, P. Yadav, **T. Patil**, P. Hegde, P. Koppolu, A. T. Weakley, S. Karia, S. Namboodiri, K. Carrion, S. Liu, S. Tomaszewski Farias, and A. Weakley. “Co-designing a Daily Management Tool to Support Individuals Living Alone with Alzheimer’s Disease.” *International Gerontechnology Conference*, Vancouver, 2026.

Research Projects

Multimodal Prediction of Alzheimer’s Disease Progression 2025
github.com/tejassvpatil/multimodal-alzheimers-prediction

- Developed a multimodal deep-learning framework using ADNI data with T1-weighted MRI, longitudinal cognitive scores, and APOE, CLU, and BIN1 genetic risk factors.
- Built 3D CNN MRI encoders, GRUs for longitudinal assessments, and MLPs for genetic features, fused through cross-modal attention.
- The tri-modal model achieved 0.918 AUC, 0.881 accuracy, 0.947 precision, and 0.818 recall for stable versus progressive mild cognitive impairment.

Zero-Shot Promptable Instance Segmentation via Repurposed Diffusion Models 2024
[project report](#)

- Repurposed a pretrained Stable Diffusion U-Net for category-agnostic instance segmentation by fine-tuning in pixel space, producing deterministic one-step inference.
- Added multi-point positive and negative prompting with similarity maps and benchmarked on COCO against Segment Anything.

Transformers for Failure Prediction in High-Performance Computing Logs 2024
github.com/allisonaustin/log-transformers

- Developed minGPT and BERT anomaly-detection models for HPC logs; implemented probabilistic inference and attention mechanisms and improved prediction accuracy by 15% on Los Alamos System 20 data.

Blood Glucose Level Prediction Using Machine Learning 2024
github.com/anirudh-r-hub/stats221project

- Predicted one-hour-ahead glucose for Type 1 diabetes using BrisT1D CGM, insulin, nutrition, steps, heart-rate, and activity data; compared Random Forest, Decision Tree-AdaBoost, LSTM, and LSTM-BiLSTM models. The best cascade model reached RMSE 0.1497 and R^2 0.7447.

Empirical Study on NPM Ecosystem Growth 2024
github.com/vaishnavikosuri/Study-on-npm-ecosystem

- Analyzed approximately 30,000 packages using npm, GitHub, Libraries.io, and Snyk data; modeled activity and dependency patterns with EDA, Pearson correlation, and Random Forest regression; validated a Threat Score metric with ANOVA ($p < 10^{-8}$).

Unveiling C++ Security: Vulnerability Detection and Mitigation in Open Source Code 2024
github.com/AmeyGohil/Damn_Vulnerable_CPP_Program

- Studied buffer overflows, use-after-free, memory leaks, double free, dangling pointers, uninitialized access, injection, integer overflows, and unsafe conversions; developed a deliberately vulnerable C++ repository for reproducible analysis.

Secure Authentication Using Zero-Knowledge Proofs 2021
[IEEE paper](#)

- Built a two-step SRP-based, verifier-backed challenge-response login; encrypted protocol parameters with AES, persisted nonce and tag, benchmarked latency, and evaluated replay, MITM, sniffing, and offline-guessing resistance.

Teaching Experience

University of California, Davis — Teaching Assistant Aug 2024–present
Computer Science, technology ethics, writing, and humanities Davis, CA

- Led discussions and office hours, graded assignments and examinations, and provided individualized feedback for ECS 188: Ethics in an Age of Technology; ECS 32A: Introduction to Programming; AMS 001C; CHI 010; and HIS 017A.
- Received a 5/5 evaluation for ECS 188.

Honors, Awards, and Certifications

-
- Outstanding Graduate Teaching Assistant Award nomination, UC Davis.
 - Rising Star Award, Capgemini, for contributions to customer-facing banking workflows associated with improved conversion rates.
 - Microsoft Certified: Azure Fundamentals (AZ-900).